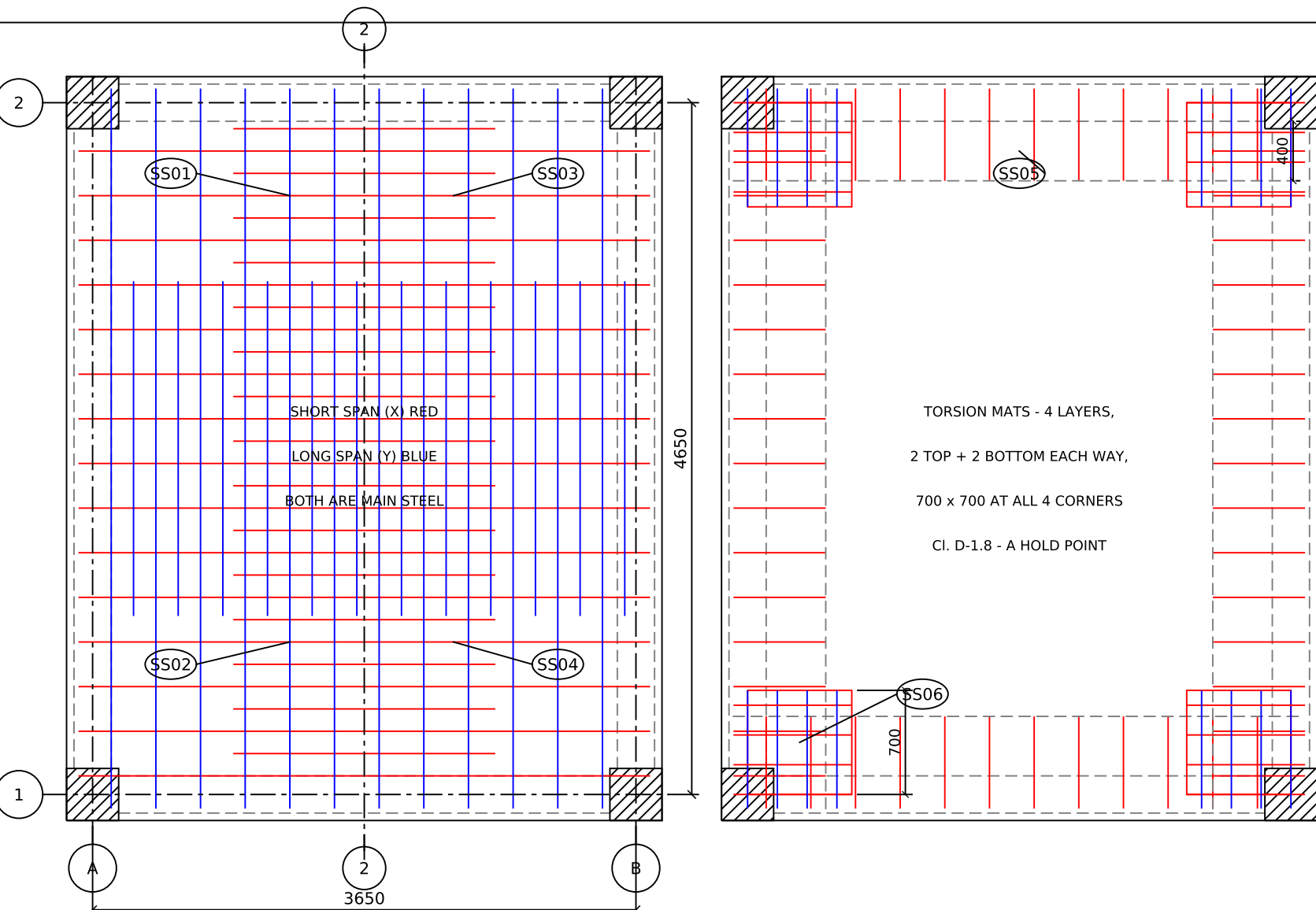
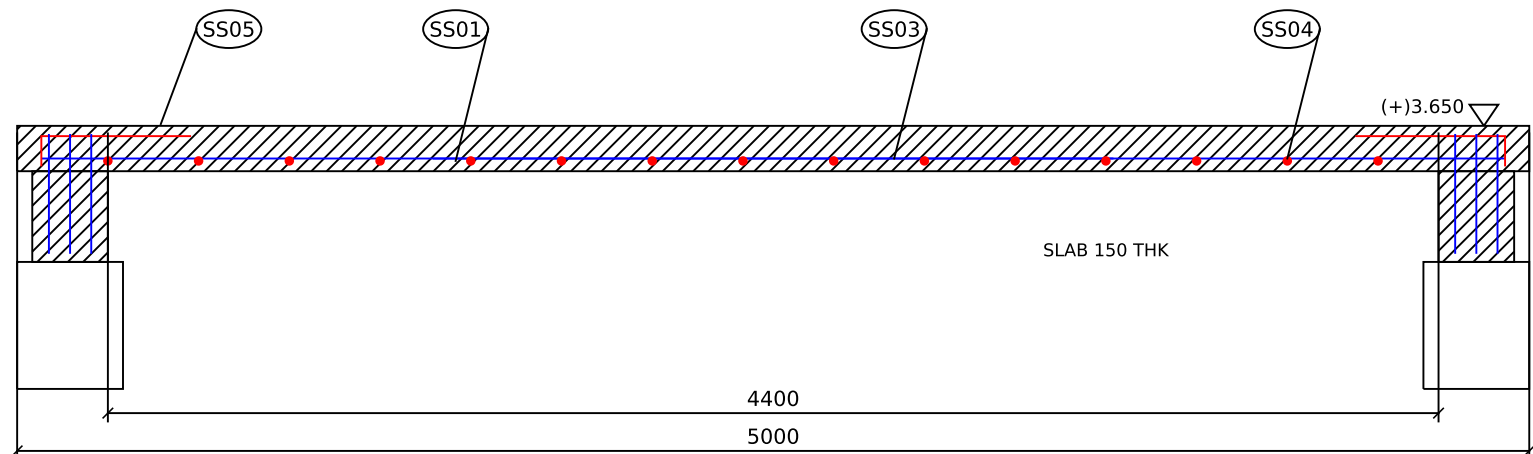


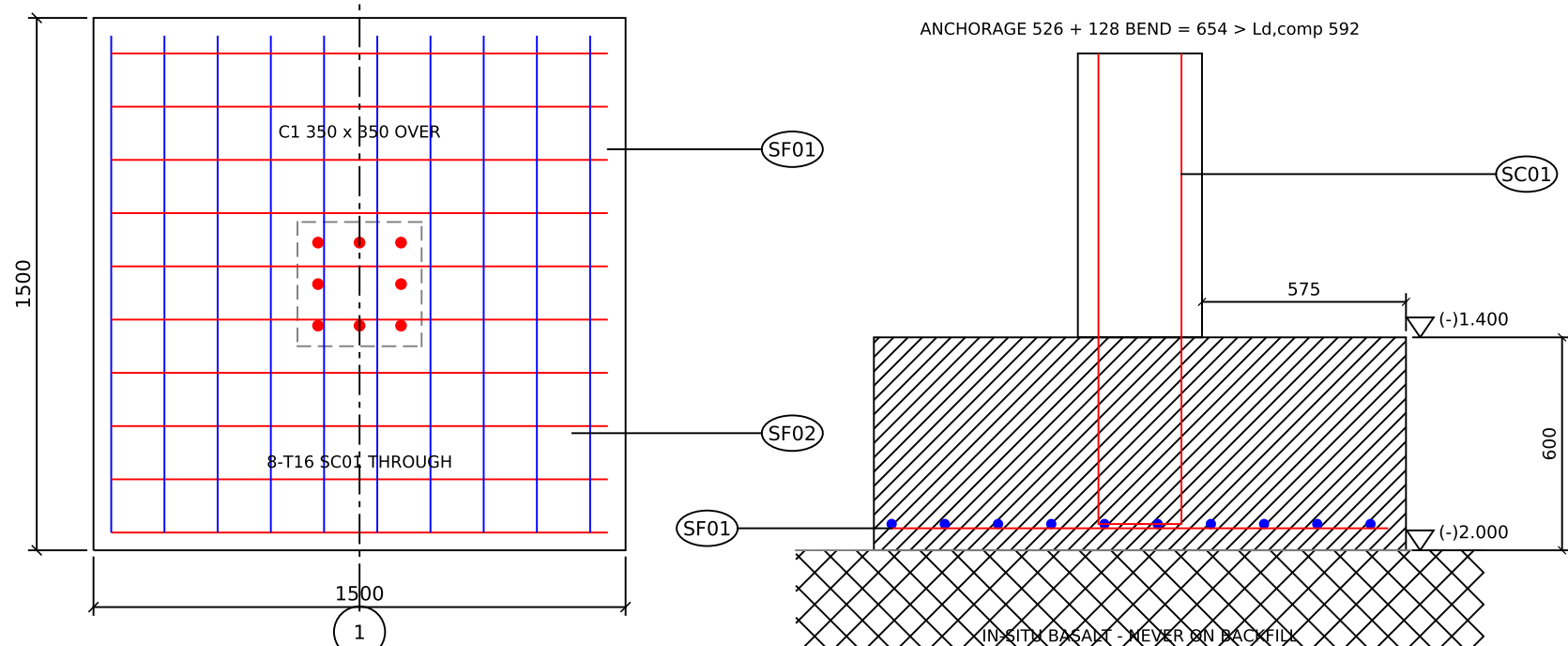


1 SLAB S1 - BOTTOM REINFORCEMENT PLAN 1 : 40 T8 @ 150 BOTH WAYS = 335 mm2/m, COVER 30

2 SLAB S1 - TOP REINFORCEMENT PLAN 1 : 40 400 EDGE BANDS (CI. D-1.6) + 700 x 700 CORNER MATS (CI. D-1.8)



3 SLAB S1 - LONGITUDINAL SECTION 2-2 1 : 25 TYPICAL AT FIRST FLOOR AND ROOF. B1 250 x 450 AT EACH END - SEE SHEET 08



4 ISOLATED FOOTING F1 - REINFORCEMENT PLAN 1 : 20 T12 @ 150 B/W BOTTOM, COVER 50, 4 No.

5 ISOLATED FOOTING F1 - SECTION 1-1 1 : 20 DEPTH SET BY STARTER ANCHORAGE

FOOTING SCHEDULE							
MARK	No.	SIZE	FOUNDING	TOP	COVER	d	REINFORCEMENT
F1	4	1500 x 1500 x 600	(-)2.000	(-)1.400	50	542	T12 @ 150 B/W BOTTOM (754)
FOUNDED ON IN-SITU BASALT. PRESUMPTIVE SBC 3240 kPa, IS 1904 TABLE 1 [ASSUMED]							
SERVICE q,max 157.6 kPa = 4.9 % OF SBC e = 0.123 < L/6 = 0.250, NO TENSION							
THE 600 DEPTH IS SET BY STARTER ANCHORAGE, NOT BY BENDING OR SHEAR							

SLAB S1 - ELEMENT SCHEDULE					
ELEMENT	THK	COVER	d x / d y	REINFORCEMENT	BASIS
S1 BOTTOM, BOTH WAYS	150	30	116 / 108	T8 @ 150 (335 mm2/m)	IS 456 Cl. 26.5.2.1 min 180
S1 TOP, DISCONTINUOUS EDGES	150	30	116	T8 @ 300, 400 INTO THE SPAN	Cl. D-1.6, all four edges
S1 CORNER TORSION MATS	150	30	-	T8 @ 200, 4 LAYERS, 700 x 700	Cl. D-1.8, all four corners
S1 OCCURS TWICE - FIRST FLOOR AND ROOF (MASTER A.4.8). THE FIRST FLOOR GOVERNS.					

BAR MARK SCHEDULE - FOOTING F1 AND SLAB S1					
MARK	BAR	SPACING	LOCATION	CUT mm	No.
SF01	T12	150	BOTTOM, BOTH WAYS - LAYER 1 (X)	1400	40
SF02	T12	150	BOTTOM, BOTH WAYS - LAYER 2 (Y)	1400	40
SS01	T8	300	BOTTOM, SHORT SPAN (X), FULL LENGTH (D4)	3840	30
SS02	T8	300	BOTTOM, SHORT SPAN (X), CURTAILED AT 0.25 L	1760	30
SS03	T8	300	BOTTOM, LONG SPAN (Y), FULL LENGTH (D4)	4840	24
SS04	T8	300	BOTTOM, LONG SPAN (Y), CURTAILED AT 0.25 L	2250	24
SS05	T8	300	TOP, 400 EDGE BAND, ALL FOUR EDGES	720	108
SS06	T8	200	CORNER TORSION MATS, 4 LAYERS, 700 x 700	795	128

IS 456 ANNEX D - TWO-WAY SLAB DETAILING		
CLAUSE	REQUIREMENT	AS DETAILED
D-1.1 / Table 26	CASE 9, FOUR EDGES DISCONTINUOUS BUT RESTRAINED	alpha x 0.0777, alpha y 0.056
Cl. 24.4	ly/lx = 4508/3516 = 1.282 < 2 -> TWO-WAY	CONFIRMED
D-1.2 / D-1.3	MIDDLE STRIP 3/4, EDGE STRIPS 1/8 EACH SIDE	TABLE 26 MOMENTS APPLY TO THE MIDDLE STRIPS ONLY
D-1.4	50 % TO WITHIN 0.1 L OF A DISCONTINUOUS EDGE	SS01 / SS03 FULL LENGTH, SS02 / SS04 CUT AT 0.25 L
D-1.6	TOP STEEL = 50 % OF MIDSPAN, 0.1 L = 400 INTO THE SPAN	SS05 T8 @ 300, ALL FOUR EDGES
D-1.7	EDGE STRIPS - MINIMUM REINFORCEMENT ONLY	T8 @ 150 THROUGHOUT COVERS IT
D-1.8	CORNER TORSION = 3/4 OF MIDSPAN, 4 LAYERS OVER lx/5 = 700	SS06 T8 @ 200 - A HOLD POINT
Cl. 26.5.2.2	MAX BAR DIA D/8 = 18.75	T8 OK
Cl. 26.3.3(b)	MAX SPACING 3d OR 300	150 BOTTOM, 300 TOP OK

UNDERGROUND CBRN-HARDENED
BLAST-RESISTANT PROTECTIVE
STRUCTURE + SENTRY POST
PUNE, MAHARASHTRA
STRUCTURAL - PHASE 2 + M1

NOTES

- ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS.
- WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
- THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH AND STR DRGS.
- SENTRY POST CONCRETE M30. REINFORCEMENT Fe500D TO IS 1786:2008. COVER 40 TO COLUMNS, 30 TO BEAMS AND SLABS, 50 TO FOOTINGS (IS 456 Cl. 26.4.2 / TABLE 16).
- FOOTINGS ARE FOUNDED ON IN-SITU BASALT AT (-)2.000, NEVER ON BACKFILL. THAT IS THE WHOLE REASON THE SENTRY POST IS SITED AT LEAST 10 m CLEAR OF THE SHELTER EXCAVATION.
- THE 600 FOOTING DEPTH IS GOVERNED BY THE ANCHORAGE OF THE COLUMN STARTERS. NOT BY BENDING OR SHEAR. IT MUST NOT BE 'OPTIMISED' TO 350.
- SLAB S1 IS A TWO-WAY SLAB ON IS 456 TABLE 26 (ANNEX D-1.1, CASE 9). TABLE 26 IS VALID ONLY BECAUSE THE CORNER TORSION MATS ARE PROVIDED. THE MATS ARE A HOLD POINT FOR REINFORCEMENT INSPECTION.
- THE SENTRY POST FRAME IS THE IS 13920:2016 ELEMENT OF THIS PROJECT. IT IS A SPECIAL MOMENT-RESISTING FRAME, R = 3.0, Ah = 0.100, DESIGNED ON THE STAAD BASE SHEAR Vb = 73.18 kN.
- THE TAGS (D3) AND (D4) IN THE BAR MARK SCHEDULE ARE DECLARED DETAILING DECISIONS, RECORDED IN MASTER PART H.40.4.
- THE OPEN ITEMS THAT AFFECT THIS SHEET - SR2-V4 THE FOOTING TOP LEVEL, SR2-V5 NO BLINDING UNDER F1, SR2-V6 CONFINING STEEL INTO THE FOOTING - ARE IN MASTER PART H.40.5.
- CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.

STRUCTURAL REINFORCEMENT DETAILING OF SENTRY POST - FOOTING & SLAB

CBRN HARDENED UG OPS ROOM

SHEET 09

Project number	01
Date	16 SEP 2026
Drawn by	SYN 01
Checked by	DR IR CHAUDHARI
STR009	
Scale	As indicated